Linear systems of three unknowns

1 Determine whether each ordered triple is a solution of the corresponding system of equations:

a Triple: $(-9, -7, 4)$
System:
$$\begin{aligned} -2x + 5y - 3z &= -29 \\ -3x + y + 4z &= 38 \\ -4x + 2y + 3z &= 34 \end{aligned}$$

b Triple: $(1, -5, -3)$
System:
$$\begin{aligned} 4x - 5y + 3z &= 20 \\ 4x - 5y - z &= 32 \\ -x + 2y - 5z &= 4 \end{aligned}$$

c Triple: $\left(\frac{3}{4}, \frac{4}{5}, \frac{2}{5}\right)$
System:
$$\begin{aligned} 5x + y + 3z &= \frac{23}{4} \\ 4x + 3y + 5z &= \frac{37}{5} \\ 2x - 4y + 3z &= -\frac{1}{2} \end{aligned}$$

d Triple: $(1.1, 0.3, 1.7)$
System:
$$\begin{aligned} 3x + y - 5z &= -5.9 \\ 5x + y - 4z &= -1 \\ 5x - 2y - 4z &= -1.9 \end{aligned}$$

e Triple: $(-8, -1, 2)$
System:
$$\begin{aligned} x + 5y + 4z &= -8 \\ 4x - y + 3z &= -25 \\ 2y - z &= -4 \end{aligned}$$

f Triple: $(-6, 3, -9)$
System:
$$\begin{aligned} x + 5y + 3z &= -18 \\ 5x + y - 3z &= -3 \\ 5x + 3y - z &= -12 \end{aligned}$$

2 Solve the following systems of equations:

a
$$\begin{aligned} x - 5y + z &= -16 \\ 2x + 5y + 2z &= 13 \\ -x - 5y - 3z &= -10 \end{aligned}$$

b
$$\begin{aligned} x + y &= 6 \\ y + z &= 13 \\ x + z &= 9 \end{aligned}$$

c
$$\begin{aligned} x + 3y &= -11 \\ 5y + 4z &= -33 \\ x + 2z &= 0 \end{aligned}$$

d
$$\begin{aligned} 5x + 2y + 4z &= -13 \\ 6y + 4z &= 22 \\ 5x - 2y + 3z &= -31 \end{aligned}$$

e
$$\begin{aligned} 3x + 2y - z &= -9 \\ 6x - 4y + z &= -20 \\ 3x + 6y + 4z &= 5 \end{aligned}$$

f
$$\begin{aligned} 3x + 3y - z &= 6 \\ 2x - 2y + z &= 3 \\ 2x + 5y - z &= 2 \end{aligned}$$

g
$$\begin{aligned} 3x + 2y - 2z &= 4 \\ y + 2z &= 5 \\ 3x - y - 2z &= 7 \end{aligned}$$

h
$$\begin{aligned} x &= 8(y - z) \\ 2z &= 9(y - x) \\ x + z &= 2y - 3 \end{aligned}$$

i
$$\begin{aligned} x &= 2(y - z) \\ y &= 4(z - x) \\ x + y &= z + 6 \end{aligned}$$

j
$$\begin{aligned} 9x + 5y - 2z &= 25 \\ x + 4y - 7z &= 30 \\ 3x - 6y + 8z &= -33 \end{aligned}$$

3 How many solutions do each of the follow[redacted]ystems of equations have?

a
$$\begin{aligned} 3x + 4y + 2z &= -1 \\ 5x - 2y - 3z &= -23 \\ -3x - 4y - 2z &= 1 \end{aligned}$$

b
$$\begin{aligned} 6x + 7y - 4z &= -13 \\ 12x + 14y - 8z &= -24 \\ 3x - 2y + 5z &= 17 \end{aligned}$$

4 Consider the system:

Equation 1	Equation 2
$2x - 3y - z = -19$	$y - z = -1$

Let $z = t$ be arbitrary.

a Solve Equation 2 for y with respect to t.

b Using your result from part a, solve Equation 1 for x with respect to t.

c Hence, write down the solution to the system with respect to the arbitrary variable t.

5 Consider the system:

Equation 1	Equation 2
$4x + 5y - z = 11$	$x + y + 2z = 9$

Let $z = t$ be arbitrary.

a Solve Equation 2 for y with respect to x and t.

b Solve Equation 1 for x with respect to t.

c Hence, solve for y with respect to t.

d Hence, write down the solution to the system with respect to the arbitrary variable t.

6 Consider the system:

Equation 1	Equation 2
$x - 3y + 3z = 6$	$x - 2y + 4z = 5$

Let $z = t$ be arbitrary.

a Solve Equation 1 for x with respect to y and t.

b Solve Equation 2 for y with respect to t.

c Hence, solve for x with respect to t.

d Hence, write down the solution to the system with respect to the arbitrary variable t.

7 Consider the following system of three equations:

$$
\begin{aligned}
3x + 5y + 8z &= 136 \\
27x + 45y + 72z &= 872 \\
x + y + z &= 25
\end{aligned}
$$

a Does the system have one solution, infinite solution or no solution?

b Is the system consistent or incosistent?

8 Consider the following system of 3 equations:

$$
\begin{aligned}
8x + 5y + 8z &= 97 \\
6x + y + 4z &= 45 \\
x + y + z &= 14
\end{aligned}
$$

a Solve the system.

b Is the system independent or dependent?

9 Consider the following system of 3 equations:

$$
\begin{aligned}
7x + 3y + 3z &= 3.2 \\
8x + 5y + 3z &= 4.4 \\
4x + 9y + 4z &= 5.7
\end{aligned}
$$

a Solve the system.

b Is the system consistent or incosistent?

10 Find a system of three linear equations in three variables that has $(-4, -2, -1)$ as a solution.

11 Solve the following systems of equations:

a
$$
\begin{aligned}
\frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= 13 \\
\frac{1}{x} + \frac{1}{y} &= 9 \\
\frac{1}{x} - \frac{1}{z} &= 2
\end{aligned}
$$

b
$$
\begin{aligned}
\frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= 11 \\
\frac{1}{y} + \frac{1}{z} &= 8 \\
\frac{1}{x} - \frac{1}{y} &= -3
\end{aligned}
$$

12 For each of the following system of 3 equa[illegible]:

i Solve the system.

ii Determine whether the system is consistent or inconsistent.

iii Determine whether the system is dependent or independent.

a
$$\begin{aligned} x-5z&=-8\\ y+4z&=16\\ 6y+2z&=30 \end{aligned}$$

b
$$\begin{aligned} x+3z&=22\\ y+8z&=49\\ 8z&=48 \end{aligned}$$

c
$$\begin{aligned} x+5y-7z&=-8\\ y+6z&=56\\ 6y+9z&=120 \end{aligned}$$

d
$$\begin{aligned} 9x+9y+6z&=303\\ 16x+18y+10z&=536\\ x+y+z&=39 \end{aligned}$$

e
$$\begin{aligned} 5x+4z&=42\\ 5x+9y+&=37\\ 6y+7z&=74 \end{aligned}$$

f
$$\begin{aligned} 9x+7y&=180\\ x+y+5z&=27\\ 4x+6y+4z&=110 \end{aligned}$$

g
$$\begin{aligned} 9x+7y+3z&=151\\ 8x+y+9z&=142\\ x+y+z&=23 \end{aligned}$$

h
$$\begin{aligned} 8x+9y+2z&=51\\ 4x+8y+9z&=101\\ 8x+4y+9z&=109 \end{aligned}$$

i
$$\begin{aligned} 3x+7y&=4.4\\ x+y+6z&=3.8\\ 5x+6y+8z&=8.5 \end{aligned}$$

Applications

13 Consider the equation: $Ax+By+Cz=12$.

a One solution to this equation is $\left(1,\dfrac{3}{4},3\right)$. Substitute these values into the equation, to give an equation in terms of A, B, and C.

b A second solution to the equation is $\left(\dfrac{4}{3},1,2\right)$. Substitute these values into the equation, to give an equation in terms of A, B, and C.

c A third solution to the equation is $(2,1,1)$. Substitute these values into the equation, to give an equation in terms of A, B, and C.

d Solve your three equations for A, B, and C.

14 The fraction $\frac{1}{24}$ can be written as the following sum:

$$\frac{1}{24} = \frac{x}{8} + \frac{y}{4} + \frac{z}{3}$$

where the numbers x, y, and z are solutions of:

$$\begin{aligned} x + y + z &= 1 ------- \text{equation } 1 \\ 2x - y + z &= 0 ------- \text{equation } 2 \\ -x + 2y + 2z &= -1 ------ \text{equation } 3 \end{aligned}$$

a Solve this system of equations for x, y, and z.

b Use your values of x, y, and z to determine whether the following equation is true or false:

$$\frac{1}{24} = \frac{x}{8} + \frac{y}{4} + \frac{z}{3}$$

15 Consider the quadratic with equation $y = ax^2 + bx + c$ that passes through the points $(1, 15)$, $(2, 5)$ and $(4, -3)$.

a Set up a system of equations in the variables a, b and c using the points given.

b Solve your equations for a, b, and c.

c Write down the quadratic equation that passes through the points $(1, 15)$, $(2, 5)$ and $(4, -3)$.

16 The equation of a circle can be written in the general form $x^2 + y^2 + ax + by + c = 0$. The circle below passes through the points $(2, -4)$, $(5, 1)$ and $(3, -3)$:

a The general form of a circle given may be rearranged to the following form: $ax + by + c = -\left(x^2 + y^2\right)$. Use this form to set up a system of equations in the variables a, b and c from the given points.

b Solve your equations for a, b, and c.

c Write down the equation of the circle that passes through the points $(2, -4)$, $(5, 1)$ and $(3, -3)$.